CSE 5315 Exam 1 – February 25, 2025

Name (as it appears on Canvas):

- You have 80 minutes to solve the following problems.
- No books, computers, phones etc. are allowed.
- Scientific calculators are permitted. Graphing calculators are not permitted.
- You may have one sheet of notes, with both sides used. Put your name on the note sheet and turn it in with your exam.
- Circle your final answers for all questions.
- For partial credit, show your work for all problems and feel free to explain your reasoning.

If you have a 2 × 2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is then the matrix inverse is $A^{-1} = \frac{\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}}{ad-bc}$

Problem 1	(20 pts)
Problem 2	(20 pts)
Problem 3	(15 pts)
Problem 4	(5 pts)
Problem 5	(20 pts)
Problem 6	(20 pts)
Total	(100)

Problem 1 (20 Points – 4 points each) : Answer the following in a two or three short sentences

a) What is the reason to use partial pivoting in Gaussian Elimination?

b) How does the Convergence rate of the Bisection method compare to that of Newton's method?

c) What are one advantage and one disadvantage of Secant method to solve root problems.

d) When might you prefer to use the Secant method over the Newton's method to find roots?

e) What is your favorite method to find the roots of nonlinear equations and why?

Problem 2 (20 Points) : Consider the system of linear equations below.

$$x_1 + x_2 = 2$$

$$1.0001x_1 + x_2 = 2.0001$$

a) Find the Forward Error of an approximate solution $\hat{x}_1 = 1$, $\hat{x}_2 = 1.0001$
b) Find the Backward Error of an approximate solution $\hat{x}_1 = 1$, $\hat{x}_2 = 1.0001$
c) Find the error magnification factor of an approximate solution $\hat{x}_1 = 1$, $\hat{x}_2 = 1.0001$
d) Find the Condition number of the coefficient matrix.
Please express your solutions to four decimal places to help the graders

Problem 3 (15 Points)

Consider the following system of equations. Perform PA=LU factorization to get the P, L and U matrices. You do not have to solve for x y and z

$$\begin{aligned} 2x + y + z &= 5 \\ 2x + 2y + 3z &= 1 \\ x + 2y + 3z &= 2 \end{aligned}$$

Problem 4 (5 points) Consider the system

$$x_1 + x_2 + x_3 = 1$$
$$4x_1 + 3x_2 - x_3 = 6$$
$$3x_1 + 5x_2 + 3x_3 = 4$$

LU decomposition was performed to get $L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}$ $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -5 \\ 0 & 0 & -10 \end{bmatrix}$.

Solve for x_1, x_2, x_3

Problem 5 (20 points) a) Apply three steps of the Bisection Method to find an approximate root for the function $f(x) = e^{-x} - x$. Start from initial bracket $[-1,1]$.

b) what is the backwards error for this solution?

Please express your solutions to four decimal places to help the graders

Problem 6 (20 points) a) Apply three steps of the Secant Method to find an approximate root for the function $f(x) = e^{-x} - x$. Start from initial conditions $x_0 = -1, x_1 = 1$.

b) what is the backwards error for this solution?

Please express your solutions to four decimal places to help the graders